# Reused Predicted Labels in Aggregates and Regression

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**Abstract**

Researchers often reuse one predicted item label across many records and construct several aggregates from those records. One mistake can then change several unit measures and estimates together. I trace these errors through totals, comparisons, and regressions using a matrix linking items to records. When an aggregate is a regression outcome with fixed observed regressors, its coefficient is a weighted sum of item labels: exact target-item confusion counts give sharp bounds, but may leave even its sign unresolved. When the aggregate is a covariate, its coefficient also depends on cross-products of labels, so linear corrections do not carry over automatically. Reuse can correlate aggregate errors across records even when errors on distinct items are independent. An existing probability audit of items with known inclusion probabilities can correct several linear estimates jointly and quantify their design uncertainty without calibrated scores. In browsing data, exact counts of 1% imposed errors leave all 84 group-comparison signs unresolved, and a 1% perturbation changes slopes when an aggregate is a covariate. Observed adult-domain classifiers agree on 97.7% of domains yet imply different visit shares. The needed validation evidence depends on where item errors enter the reported estimate.

Researchers classify websites, merchants, documents, and other items, then join their predicted labels to records of people, places, or periods. A domain label, for example, may enter the browsing histories of many people and the visit shares reported for several groups. Classification accuracy counts the domain once. Its effect on a group difference depends on how often each group visits it. An error on a widely visited domain can move several group means together, while an error on a less visited but unevenly used domain can change their difference. Error counts alone omit those locations.

The same issue reaches regression in two ways. If a label-derived aggregate is the outcome, each fixed-design coefficient is a weighted sum of the item labels. If the aggregate is a covariate, the coefficient depends on both a linear numerator and a quadratic denominator in those labels. An item error can enter many regression rows, and its effect on a coefficient need not look like ordinary attenuation. When several estimates use the same predictions, their errors can move together or cancel in a contrast. These are consequences of the links between items and records, even when errors on distinct items are independent.

Prior work treats important parts of this problem. Quantification estimates class prevalence, and cost quantification allows nonnegative item weights (Forman, 2008; Hopkins and King, 2010; Vaz et al., 2019). Predicted regression covariates and indices built by classification and aggregation are studied by Fong and Tyler (2021) and Battaglia et al. (2025); design-based supervised learning uses verified labels to correct downstream analyses (Egami et al., 2023). Shared item errors have a covariance structure familiar from shift-share designs (Adão et al., 2019). The question here is what a particular validation result establishes when one item prediction is reused across records

*gsood07@gmail.com. The observation that errors on popular domains can distort browsing totals appears in Sood (2018) and Laohaprapanon and Sood (2020). This paper develops its implications for aggregates and validation. Code: `https://github.com/finite-sample/total_error`.

and across reported estimates. Its answer depends on the downstream quantity and on which error locations validation preserves.

I represent the links by a matrix mapping item labels into record-level aggregates. For linear estimates, this yields item weights that determine the effect of each mistake. I derive sharp bounds given exact false-positive and false-negative counts for the target items, then show when those counts suffice. A probability audit of items can instead correct several linear estimates jointly, with a design covariance that retains their shared error. The regression-covariate case exposes the limit of those linear results: its denominator contains item cross-products. None of the fixed-label bounds or probability-audit results requires calibrated scores.

A small example shows how one item mistake changes several records and both roles of an aggregate in regression. The empirical analysis uses an observed person–domain matrix to show how reuse and skewed visits affect uncertainty and count-based bounds under imposed errors. A second analysis compares observed classifications of adult domains and their visit-weighted shares. The latter measures disagreement between classifiers, without treating either as verified truth.

## 1 From labels to estimates

Let $y_j \in \{0,1\}$ be the reference label of item $j$, for $j = 1, \ldots, k$, and let $q_j \in \{0,1\}$ be its prediction. A known matrix $C = (c_{ij}) \in \mathbb{R}^{n \times k}$ records how item $j$ contributes to unit $i$. An item can enter many units, and a unit can contain many items. The measures built from reference and predicted labels are $T = Cy$ and $S = Cq$. A reported statistic $F(T, Z)$, with other observed inputs $Z$, incurs error $F(Cq, Z) - F(Cy, Z)$. The rows of $C$ can also stack several aggregations made from the same predictions. For fixed unit weights $a$, a linear target has item weights

$$\theta = a^{\mathsf{T}} T = w^{\mathsf{T}} y = \sum_{j=1}^{k} w_j y_j, \qquad w = C^{\mathsf{T}} a. \tag{1}$$

Totals have nonnegative weights; group comparisons have signed weights. Known-denominator proportions are also linear after the denominator is absorbed into $C$. Different targets may use different matrices and unit weights while reusing the same $q$. A label-dependent denominator or a regression that uses $Cq$ as a covariate is nonlinear in the labels (Section 2).

In the browsing application, $c_{ij}$ counts person $i$'s visits to domain $j$. For visit shares, we divide each row of $C$ by that person's observed total visits. These denominators do not depend on the domain labels.

Let $e = q - y$. The prediction-based estimate is $\widehat{\theta} = w^{\mathsf{T}} q$, so its error is exactly

$$\Delta = \widehat{\theta} - \theta = w^{\mathsf{T}} e. \tag{2}$$

The weights determine which mistakes matter. A popular item can dominate a total but cancel from a comparison if both groups use it equally. For several estimates, let $W = [w^{(1)}, \ldots, w^{(R)}]$. Their joint error is $W^{\mathsf{T}} e$: the same item mistake can move several estimates together. For realized labels and predictions these errors are fixed. Under a specified repeated-prediction process, with $\Sigma = \mathrm{Cov}(e)$, the record-level errors $d = Ce$ have covariance $C\Sigma C^{\mathsf{T}}$ and the estimate errors have covariance $W^{\mathsf{T}}\Sigma W$. If distinct item errors are independent, $\mathrm{Cov}(d_i, d_\ell) = \sum_j c_{ij} c_{\ell j} \mathrm{Var}(e_j)$: reuse alone can link errors across records. The bounds and audit below instead condition on realized predictions. Appendix D develops the covariance result.

For illustration, consider three records and two items:

$$C = \begin{pmatrix} 4 & 0 \\ 1 & 2 \\ 0 & 1 \end{pmatrix}, \qquad y = (0,1)^\mathsf{T}, \qquad q = (1,1)^\mathsf{T}.$$

One false positive changes the record measures from $T = (0,2,1)^\mathsf{T}$ to $S = (4,3,1)^\mathsf{T}$: it affects two records unequally. With an observed variable $Z = (0,1,2)^\mathsf{T}$ and an intercept, the slope from regressing $T$ on $Z$ is $1/2$; using $S$ as the outcome gives $-3/2$. Regressing $Z$ on $T$ also gives $1/2$, while using $S$ as the covariate gives $-9/14$. The two regression roles have different error formulas.

## 2 Aggregates in regression

An aggregate built from predicted labels can be either a regression outcome or a covariate. If it is the outcome and $X$ is a fully observed, full-rank design, the complete-label least-squares coefficients are $\beta^* = ACy$, where $A = (X^\mathsf{T}X)^{-1}X^\mathsf{T}$. The prediction-based coefficients are $ACq$. A contrast $r^\mathsf{T}\beta^*$ has item weights $w = C^\mathsf{T}A^\mathsf{T}r$, so its error is $w^\mathsf{T}e$. The count bounds and item-audit correction below apply to this contrast. The coefficients are descriptive for the observed frame; causal or population interpretations require additional assumptions.

If the aggregate is a covariate, the coefficient is nonlinear in the labels. The same is true of a proportion whose denominator depends on predicted labels. The linear bounds below do not apply to these quantities without evaluating their full downstream map.

For example, let $Y$ be an observed outcome and $V$ a full-rank matrix of fully observed controls, including an intercept. The coefficient that ordinary least squares would report with complete item labels is

$$\beta(y) = \frac{T^\mathsf{T}M_V Y}{T^\mathsf{T}M_V T} = \frac{y^\mathsf{T}C^\mathsf{T}M_V Y}{y^\mathsf{T}C^\mathsf{T}M_V Cy}, \qquad M_V = I - V(V^\mathsf{T}V)^{-1}V^\mathsf{T}, \tag{3}$$

when the denominator is positive. Both numerator and denominator change when labels change, so ordinary attenuation is not a general conclusion. The exact error identity and item-audit moment formulas are in Appendix C. Those formulas do not provide an interval for $\beta(y)$: a ratio of estimated moments can be biased, unstable, or undefined.

Fong and Tyler (2021) study regression with predicted covariates, and Battaglia et al. (2025) include indices built by classification and aggregation. This paper's item-to-record map makes the unit of validation explicit: verifying item $j$ may correct pieces of many regression rows, without revealing any row's complete covariate. Work using validated complete regression rows, including Kluger et al. (2025), requires a different sampling unit or an adaptation here. Overlap in $C$ has the common-shock structure of shift-share designs (Adão et al., 2019). It can matter in either direction for a coefficient. An item with the same weight in every unit affects their totals, but its error disappears from a regression with an intercept because $M_V C_{\cdot j} = 0$.

There is also a generated-regressor issue beyond the covariance of the rows. The OLS numerator is linear in the labels but its denominator contains $G = C^\mathsf{T}M_V C$, including cross-products of items that enter the same units after residualizing on controls. Appendix C gives the exact plug-in score and its expectation under a specified prediction-error process. If the same predictions build two or more covariates, their measurement errors also enter the off-diagonal cross-products of the regression's normal equations. In that case there need not be a single attenuation factor: an error shared by two covariates can move both coefficients. The relevant cross-products are given in Appendix C. A verified item can inform many units' aggregates. For equation (3), an item audit

must account for item pairs in the quadratic denominator; correcting each aggregate separately is not enough.

A model-based target requires different assumptions. If information $\mathcal{H}$ contains $C, V$ and the prediction inputs, and $p_j = \mathrm{E}[y_j \mid \mathcal{H}]$, then $Cp = \mathrm{E}[T \mid \mathcal{H}]$. Under $Y = V\gamma + \beta T + \varepsilon$ with $\mathrm{E}[\varepsilon \mid \mathcal{H}] = 0$ and a full-rank design, regressing $Y$ on $(V, Cp)$ identifies the population $\beta$. Marginal score calibration across items does not establish these conditional means or the exclusion assumption. This population coefficient is also distinct from the fixed-sample complete-label coefficient in equation (3).

## 3 What a confusion matrix identifies

A confusion matrix can be exactly correct for the target items and still fail to determine a downstream estimate. This is an information problem before it is a sampling problem. Fix the predictions $q$ and the item weights $w$ in equation (1). We now ask what a validation summary establishes about the fixed target $\theta = w^{\mathsf{T}} y$, without specifying a stochastic model for how the classifier's mistakes were generated.

Let $P = \{j : q_j = 1\}$ and $N = \{j : q_j = 0\}$. Suppose the entire target-population confusion matrix is known: $f$ predictions in $P$ are false positives and $g$ predictions in $N$ are false negatives. This grants more information than ordinary held-out validation, whose error rates require a transport assumption. For a finite collection $B$ of weights, let $S_-(B, m)$ and $S_+(B, m)$ denote the sums of its $m$ smallest and largest elements, with a zero sum when $m = 0$.

**Proposition 1** (Sharp bounds from confusion counts)**.** *Across binary label vectors compatible with $q, f, g$, the smallest and largest possible errors $\Delta = \widehat{\theta} - \theta$ are*

$$\begin{aligned} L &= S_-(w_P, f) - S_+(w_N, g), \\ U &= S_+(w_P, f) - S_-(w_N, g). \end{aligned} \tag{4}$$

*Consequently $[\widehat{\theta} - U, \widehat{\theta} - L]$ is the smallest interval containing every compatible value of $\theta$. Both endpoints are attainable.*

*Proof.* Only false positives and false negatives contribute: $\Delta = \sum_{j \in \mathrm{FP}} w_j - \sum_{j \in \mathrm{FN}} w_j$. Select $f$ indices from $P$ and $g$ from $N$. The two selections have disjoint supports and no other restrictions. Selecting the smallest positive-stratum weights and largest negative-stratum weights minimizes the error; reversing these selections maximizes it. Assign the selected indices the corresponding true labels to attain each endpoint. □

The compatible values form a finite set, so the interval is its convex hull; not every interior value need be attainable. A reported sign is established by these counts only if the bounding interval excludes zero. The calculation requires sorting, rather than a general integer-programming solver.

For example, take $q = (1, 1, 0, 0)$, $w = (1, -1, 1, -1)$, and $f = g = 1$. The same confusion matrix permits errors of $-2$, 0, and 2. Prevalence, accuracy, FPR, and FNR are identical across those assignments. Adding correctly classified items with zero weight makes accuracy arbitrarily close to one without resolving the target. In the browsing application, these can be domains that affect individual exposure but cancel from a group contrast.

**Proposition 2** (When counts suffice)**.** *In a stratum of $M$ items sharing the same predicted label, with exactly $m$ errors, the error sum is point identified if and only if $m \in \{0, M\}$ or all its contrast weights are equal. Its identification width obeys*

$$S_+(w, m) - S_-(w, m) \leq \min(m, M - m)\{\max_j w_j - \min_j w_j\}. \tag{5}$$

*Widths add across strata. For several comparisons, counts determine every comparison if and only if their vector of item weights is constant within each stratum with an interior error count.*

*Proof.* Constant weights, no errors, and all errors each determine the sum. If $0 < m < M$ and two weights differ, choose an error set containing one of those indices but not the other. Exchanging them changes the sum, proving necessity. The difference between two sums of $m$ terms is at most $m$ times the range. Taking complements gives the corresponding bound with $M - m$ terms. The vector statement applies the exchange argument to each coordinate. □

For a stratum mixing predicted labels, the corresponding criterion applies to the signed contributions $(2q_j - 1)w_j$, not to $w_j$ alone. The result specifies what a finer validation table must preserve. Stratifying by predicted label alone is sufficient only when the contrast attaches equal weights within each partially erroneous stratum. Partitioning items by similar contrast weights can reduce ambiguity; bins chosen for one comparison need not work for another. In browsing, error rates conditional on user demographics are not automatically available: the same domain can enter many groups.

## 4 What collected validation labels can support

An existing probability sample of verified target items can recover information that a confusion matrix discards; it need not have been allocated optimally. The construction uses established finite-population counting and audit ideas (Stark, 2008; Shekhar et al., 2023). Partition the target items using information known before labels are audited, then draw a fixed-size random sample without replacement within each stratum. Exact hypergeometric inversion gives a confidence interval $[\ell_h, u_h]$ for the number of incorrect predictions in stratum $h$; allocate error probabilities across strata so their sum is at most $\delta$.

Let $O$ contain the audited items and $\mathcal{Y}_A$ the binary label vectors that match their verified labels and whose error counts in each stratum lie in $[\ell_h, u_h]$. Its construction and the sorting formula for its extremes are in Appendix E.

**Proposition 3** (A simultaneous count certificate)**.** *For any fixed item weights $w$, the interval $[\min_{z\in\mathcal{Y}_A} w^\mathsf{T} z, \max_{z\in\mathcal{Y}_A} w^\mathsf{T} z]$ covers the fixed target $w^\mathsf{T} y$ with probability at least $1-\delta$ over the audit design. The same event covers every linear contrast of this one label vector. Its endpoints are sharp given the audited labels and the count intervals.*

The certificate is simultaneous because the count event constrains the same label vector for every $w$. It can nevertheless be wide: it retains error counts but discards the weights attached to unaudited error locations. The example and full audit comparison in Appendix E show this loss. Those simulations evaluate a design chosen for one coefficient; they do not establish efficient simultaneous auditing for other contrasts.

For one declared contrast, a stratified difference estimator instead keeps the audited item identities and their weighted residuals:

$$\widetilde{\theta} = \widehat{\theta} + \sum_h \frac{M_h}{n_h} \sum_{j\in O_h} w_j(y_j - q_j). \tag{6}$$

With fixed predictions and verified labels, this estimator is design unbiased. A normal interval needs a separate approximation. The conservative finite-sample Hoeffding interval in the appendix is wide in this application; the paper does not establish an efficient finite-sample interval or an optimal

audit allocation. The distinction here is between preserving the locations of audited errors and reducing them to counts.

The same collected labels can correct several estimates jointly. Let $W$ contain their item-weight vectors, let $R_j$ indicate that item $j$ was verified, and let $\pi_j > 0$ be its known inclusion probability. Set $g_j = (y_j - q_j)W_{j\cdot}^{\mathsf{T}}$. With predictions frozen before the validation sample was selected,

$$\widetilde{\boldsymbol{\theta}} = W^{\mathsf{T}}q + \sum_j \frac{R_j}{\pi_j} g_j, \qquad \mathrm{E}_R[\widetilde{\boldsymbol{\theta}}] = W^{\mathsf{T}}y. \tag{7}$$

The reusable item, not each downstream record, is the sampling unit. If $\pi_{j\ell} = \Pr_R(R_j = R_\ell = 1)$ and $\pi_{jj} = \pi_j$, the exact joint design covariance is

$$\mathrm{Cov}_R(\widetilde{\boldsymbol{\theta}}) = \sum_{j,\ell} \frac{\pi_{j\ell} - \pi_j\pi_\ell}{\pi_j\pi_\ell} g_j g_\ell^{\mathsf{T}}. \tag{8}$$

If every relevant $\pi_{j\ell} > 0$, replacing each summand by $R_jR_\ell/\pi_{j\ell}$ times that summand gives an unbiased covariance estimator. For independent Bernoulli item selection, only the diagonal terms remain, giving $\sum_j (1-\pi_j)(y_j - q_j)^2 W_{j\cdot}^{\mathsf{T}} W_{j\cdot}/\pi_j$. The off-diagonal entries of this covariance describe how a single verified item corrects several reported estimates together. Highly influential items with small $\pi_j$ can leave a valid estimator with a very unstable variance estimate and a poor normal approximation.

If validation items were chosen purposively and their inclusion probabilities are unknown, these design claims do not apply. One can instead state a model for how validation errors transfer to the target items, or report bounds under explicit restrictions. A confusion matrix from a convenience set does not identify either the target-item correction or its standard error. The ordinary regression standard error conditional on frozen predictions answers a different question; it does not include the error from unverified labels. If the same selected labels were used to fit the predictions, the frozen-proxy design argument also needs modification; cross-fitting or joint analysis of training and validation would require its own justification.

## 5 What calibrated scores would add

The bounds from exact target-item confusion counts and the probability audit above condition on fixed predictions and labels. Neither requires calibrated classifier scores. The audit needs a known probability-sampling design for target items and verified labels for the sampled items. If the error counts come instead from a separate validation collection, applying them to the target items requires a transport assumption, whether or not that collection's scores are calibrated.

A score-based plug-in uses a different source of information. Let $p_j \in [0,1]$ be a score intended to represent the probability that target item $j$ has label one, and put $\theta_p = w^{\mathsf{T}}p$. To interpret this as a conditional mean, introduce a model in which the labels are random given information $\mathcal{I}$ containing the scores and the target weights. Write $\mu_j = \mathrm{E}[y_j \mid \mathcal{I}]$. Then

$$\mathrm{E}[\theta_p - \theta \mid \mathcal{I}] = w^{\mathsf{T}}(p - \mu). \tag{9}$$

The plug-in equals $\mathrm{E}[\theta \mid \mathcal{I}]$ if and only if this weighted error is zero. A sufficient condition is $p_j = \Pr(y_j = 1 \mid \mathcal{I})$ for each target item. For a particular target, offsetting score errors can also make the weighted sum zero without item-level conditional calibration.

Ordinary calibration, $\Pr(y = 1 \mid p = s) = s$ across items, is weaker. For example, let half the items have label one and weight one, and the other half have label zero and weight zero. Give every

item score 1/2. The scores are empirically calibrated in the unweighted item collection, but $\theta_p$ is only half of the realized target. Conditioning on features that distinguish the two weight groups, or directly checking the weighted residual, exposes the failure. Calibration assessed on a separate collection also has to transfer to the target items and their label definition.

For a browsing visit share, $w_j = v_j / \sum_\ell v_\ell$ gives $\theta_p = \sum_j v_j p_j / \sum_j v_j$. Conditional probabilities for the specified content label would make this the model's expected visit share. A group contrast or regression coefficient has different, possibly signed weights; its required weighted score condition changes accordingly.

Even valid target-item conditional probabilities identify a conditional expectation, not the realized finite-item target. Under the conditional-label model, uncertainty around that expectation depends on the joint distribution:

$$\mathrm{Var}(\theta \mid \mathcal{I}) = w^{\mathsf{T}} \mathrm{Cov}(y \mid \mathcal{I}) w. \tag{10}$$

Item-level calibration supplies neither the off-diagonal covariances nor, by itself, a coverage guarantee for an interval about the realized target. Estimating the scores introduces further uncertainty. These model-based claims are distinct from the fixed-label, design-based coverage of the audit.

## 6 Reuse and error locations in browsing data

Domain labels are reused across people, and visits are concentrated on a few domains. The June 2022 YouGov Pulse/RealityMine data contain 1,132 pseudonymized users, 34,078 domains, 103,853 user–domain records, and 4,767,099 visits. The most widely used domain appears in 1,080 users' records. One domain accounts for 20.7% of visits; the most visited 1% of domains account for 83.4%. We use Blacklight scanner outcomes and include domains with recorded scanner labels. Each user's visit shares use visits to those domains as the denominator. The scanner outcomes serve as reference labels for this exercise, not error-free measures of all tracking activity. The exercise makes no causal or population-representative claim about demographic differences.

We use an intercept and indicators for gender, race, education, and age groups. Omitted categories are Male, White, HS or Below, and under 25. For each of seven scanner outcomes, we calculate all twelve non-intercept least-squares coefficients for both visit totals and visit shares. Each coefficient is a fixed weighted label sum. Two scanner columns are counts and are dichotomized at a positive value; the other five are binary.

Predictions are imposed, rather than produced by a fitted classifier. Within each true class, we randomly flip the integer part of 1%, 5%, or 10% of the labels. We then perform two distinct exercises. First, holding those predictions fixed, Proposition 1 gives the true-coefficient range compatible with their exact confusion counts. Second, holding the reference labels fixed, we rearrange the same numbers of false positives and false negatives to minimize or maximize the predicted coefficient. The latter asks whether two classifiers with identical confusion matrices can report opposing comparisons. It is a worst-case sensitivity analysis, not an estimate of likely classifier bias. Extrema are computed separately for each coefficient; a single error arrangement need not reverse all comparisons at once.

At the 1% setting, exact confusion counts do not identify the sign of any of the 84 visit-share coefficients (Table 1).

Table 1: What exact confusion counts establish at 1% imposed error in each true class

| Scanner outcome | Sign identified | Median realized error (pp) | Median bound width (pp) |
|---|---|---|---|
| Advertising trackers | 0/12 | 0.43 | 47.8 |
| Third-party cookies | 0/12 | 0.09 | 52.5 |
| Canvas fingerprinting | 0/12 | 0.34 | 29.5 |
| Session recording | 0/12 | 0.40 | 31.6 |
| Key logging | 0/12 | 2.10 | 21.3 |
| Facebook pixel | 0/12 | 0.10 | 43.6 |
| Google Analytics | 0/12 | 0.33 | 21.6 |

Each row covers twelve demographic coefficients of visit shares in the browsing data. "Sign identified" counts bounding intervals excluding zero for imposed random predictions. Realized error is the median absolute difference between the prediction-based and reference coefficients. Bound width is the median identification-interval width; both are in percentage points (pp). Error counts are rounded down within each true class. These are not sampling intervals.

The realized errors from random placement are usually much smaller than the count-compatible ranges. For example, the reference female coefficient for Facebook-pixel visit share is 0.828 percentage points. The fixed random predictions differ by 0.040 percentage points, while their count-compatible interval for the reference coefficient is $[-17.00, 16.72]$ percentage points. Moving the same numbers of errors can reverse all 84 share-coefficient signs, but those placements are adversarial. These results show what the confusion counts fail to rule out for this frame, not what a fitted classifier is likely to do.

The same browsing matrix gives a regression-covariate illustration. For each other scanner outcome, we regress its user-level visit share on the advertising-tracker visit share and the demographic indicators above. We hold the outcome labels fixed and replace only the advertising-tracker labels with the 1% random-error draw. The third-party-cookie slope changes from 0.682 to 0.622; across the six outcomes, slopes can rise or fall (Table 2). These are descriptive coefficients under one imposed draw, not estimated effects of an observed classifier.

Table 2: Imposed domain-label errors in an aggregate regression covariate

| Outcome | Recorded labels | Imposed errors |
|---|---|---|
| Third-party cookies | 0.682 | 0.622 |
| Canvas fingerprinting | 0.089 | 0.060 |
| Session recording | 0.048 | 0.052 |
| Key logging | 0.063 | 0.063 |
| Facebook pixel | 0.140 | 0.113 |
| Google Analytics | 0.016 | -0.010 |

Each row regresses the listed scanner outcome's user-level visit share on advertising-tracker visit share and demographic indicators. "Recorded labels" uses the scanner's advertising-tracker labels; "Imposed errors" flips the integer part of 1% of those domain labels within each true class in one random draw. Outcome labels and controls are held fixed. Coefficients are descriptive; the imposed errors are not an evaluation of a fitted classifier.

**How much does domain reuse matter for uncertainty?** The incidence matrix permits a separate diagnostic without fitting a classifier. Suppose each domain's prediction error is independent of the others and has the same variance, as under independent binary flips with an equal false-positive and false-negative probability. Equation (D9) compares the prediction-error variance of a linear user statistic with a calculation that keeps each user's marginal error variance but discards covariance between users. For the mean visit share the variance ratio is 323.4, and one domain accounts for

81.2% of the item-level variance under this model. The female and Black share-coefficient ratios are 2.12 and 0.77: shared domain errors can cancel in signed contrasts. These ratios describe the observed weights under stipulated errors, not uncertainty for a fitted classifier. More directly, the same predicted domain labels enter the female and male mean visit shares. Their prediction errors have correlation 0.987 under the equal-variance item model. The error variance of the female-minus-male difference is only 1.36% of a calculation that treats the two group-mean errors as independent. Skew in the domain weights produces strong common movement in the two means, while differencing removes much of it.

**An audit can retain more.** Appendix E evaluates count projection and direct weighted residual estimation on the same probability-audit draws for one declared coefficient. The count certificate covered that coefficient in every draw but never settled its sign. The direct normal intervals were narrower but undercovered in the smallest random design; the finite-sample Hoeffding baseline was wider than count projection. The exercise illustrates the information lost by counts and the limits of these particular intervals. It does not compare efficient finite-sample procedures or assess the theorem's simultaneous utility across the other coefficients.

## 7 Observed disagreement between domain classifiers

The imposed-error experiment controls the number and locations of mistakes. A separate comparison uses observed outputs from the adult-domain study of Sood and Shen (2024). Its public domain summary records 6,236,834 visits to 64,074 domains in the June 2022 YouGov panel. The study used a Piedomains rule that marks a domain when its predicted class is "porn" and its reported class probability is at least 0.033. We compare that rule with recorded "porn" categories from Bitdefender, "sex" from Forcepoint, and "Pornography" from alphaMountain. Each row retains only domains with a recorded category from that provider, and visit shares use the visits in that row's frame as their denominator. Missing provider categories are not negative labels.

Table 3: Observed classifier disagreement and visit-weighted shares

| Reference | Domains | Visits (%) | Agree (%) | Domain + (%) | Visit + (%) | Count range (%) |
|---|---|---|---|---|---|---|
| Bitdefender | 43,391 | 84.2 | 97.7 | 1.92/3.36 | 1.05/1.71 | [0.07, 82.11] |
| Forcepoint | 47,950 | 87.5 | 98.0 | 1.65/3.04 | 1.02/1.64 | [0.07, 80.61] |
| alphaMountain | 36,063 | 52.7 | 99.0 | 0.98/1.35 | 1.38/2.16 | [0.04, 68.07] |

Piedomains is the proxy in every row; the named provider supplies the reference category. "Visits" is the share of all 6,236,834 visits represented by domains with a recorded reference category. Agreement counts domains equally. "Domain +" and "Visit +" list Piedomains/provider percentages, first for domains counted equally and then for visits within that row's frame. The count range is the sharp range of the provider visit share if only the two directional disagreement counts are kept, holding Piedomains labels and visit weights fixed. The providers use different category definitions; none is an adjudicated truth label. A recorded category other than the named positive one counts as negative for this comparison, including nonspecific categories such as "misc."

On the 43,391 domains categorized by Bitdefender, which account for 84.2% of visits, the two classifications agree on 97.7% of domains. Piedomains marks 1.92% of domains positive, compared with 3.36% under Bitdefender's category. Their visit-weighted shares are 1.05% and 1.71%, respectively. The discrepancy is visible before weighting domains by visits, so this comparison does not isolate an effect of visit concentration. There are 193 Piedomains-only and 816 Bitdefender-only positive domains.

With both classifications observed, the provider's visit share is known. If the disagreement locations were discarded and only their directional counts retained, its compatible range would be [0.07, 82.11]%. A single Piedomains-negative domain carries 18.8% of visits in this frame and can dominate the count-only upper bound. Appendix F shows how that range changes when the observed label pairs of the most-visited domains are retained. This isolates information discarded by a confusion table; it does not establish that the extreme allocations are plausible classifier errors.

This is a comparison between imperfect classification systems, not an estimate of either system's error against verified content labels. The available domain-level data do not include the person–domain matrix needed to apply the same calculation to the adult study's partisan coefficients. A probability audit of domains in that matrix, with an explicit content-labeling rule, would support a direct validation analysis of those coefficients without assuming calibrated scores. The reported Piedomains class probability and the provider categories alone do not establish the conditional probabilities required for the score plug-in in Section 5.

## 8 Implications for validation

After data collection, retain the identity of each verified item and how it was selected. For declared linear targets, known item inclusion probabilities support a joint correction and design covariance even when the validation allocation was inefficient. Exact confusion counts instead constrain the number of mistakes but generally not their weighted sum; their sharp bounds state what those counts alone establish. If several targets matter, their different weight vectors determine which error locations the validation data must preserve. Heavy item influence can make the covariance estimate unstable and a normal interval unreliable even when the design formula is unbiased.

These fixed-label procedures need verified labels and, for an audit, a known sampling design. They do not require calibrated scores. Applying error rates from another collection requires transport assumptions; interpreting scores as conditional probabilities requires stronger conditions described in Section 5. For aggregated predicted covariates, estimating their components does not by itself give valid inference for a regression slope: numerator and denominator moments, their joint uncertainty, and a nonvanishing denominator all matter. The browsing-specific common-rate correction and its assumptions appear in Appendix A.

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# Appendix

## A Bias and the denominators of error rates

Let $\alpha_j$ be the probability of predicting one for a truly negative domain and $\beta_j$ the probability of predicting zero for a truly positive domain, conditional on the browsing design. Then

$$b_j = \alpha_j(1 - y_j) - \beta_j y_j, \qquad \mathrm{E}[d_i \mid C, y] = \sum_j c_{ij}\{\alpha_j(1 - y_j) - \beta_j y_j\}. \tag{A1}$$

False-positive rates condition on true negatives; false-negative rates condition on true positives. Multiplying a false-positive rate by predicted positives uses the wrong denominator. Such a calculation would instead require the probability of a true negative conditional on a positive prediction, together with assumptions that justify transferring that probability to the weighted population. These conditional probabilities are not interchangeable.

If $\alpha_j = \alpha$ on negative domains and $\beta_j = \beta$ on positive domains,

$$\mathrm{E}[d_i \mid C, y] = \alpha(V_i - T_i) - \beta T_i = V_i\{\alpha - (\alpha + \beta)\pi_i\}, \qquad \pi_i = T_i/V_i, \tag{A2}$$

where the share $\pi_i$ is defined for $V_i > 0$. At a fixed true share, expected bias scales with browsing volume unless the expression in braces is zero. For $\alpha + \beta > 0$, cancellation occurs at $\pi_i = \alpha/(\alpha + \beta)$. Equal positive error rates cancel only when half of visits are to truly positive domains. Thus the difference between FPR and FNR alone does not determine bias. Nor does the number of distinct visited domains: repeated visits matter through $c_{ij}$.

The common-rate assumption is stronger than knowing an average error rate on a validation set of domains. Errors may depend on popularity, domain type, or features associated with a particular group's browsing. In that case, equation (A1), with rates conditional on those features, is the relevant expression. The bias in a share is $\mathrm{E}[d_i \mid C, y]/V_i$; dividing by volume removes the mechanical scale factor, not misclassification bias.

With known $\alpha, \beta$ and $\alpha + \beta \neq 1$, this model gives an unbiased correction:

$$\widetilde{T}_i = \frac{\widehat{T}_i - \alpha V_i}{1 - \alpha - \beta}, \qquad \mathrm{E}[\widetilde{T}_i \mid C, y] = T_i. \tag{A3}$$

The correction is unstable when the denominator is close to zero and may return values outside $[0, V_i]$. Clipping changes its expectation. Estimating error rates introduces additional uncertainty shared across users; rates estimated on unweighted domains need not transfer to weighted exposure. This is the connection to classifier-based quantification (Vaz et al., 2019), with the additional requirement that the rates apply to each user's weights.

## B A connection to label sampling

The same mapping supplies a remedy when a researcher can verify a probability sample of items. The companion `fewlab` project develops sampling designs for reusable labels (Sood, 2026). In its browsing application, the regression decomposition is $H = AC$, with column $h_j$ giving the influence of domain $j$ on the coefficient. For exposure shares, replace $C$ by its row-normalized version. For any scalar target in equation (2), the corresponding item influence is $w_j$.

Freeze predictions $q_j$ before drawing a label sample. Let $R_j$ indicate selection, with known inclusion probability $s_j > 0$ for every item. Conditional on the fixed population and predictions,

the difference estimator is

$$\widetilde{\theta} = \sum_j w_j q_j + \sum_j \frac{R_j}{s_j} w_j (y_j - q_j). \tag{B1}$$

Since $\mathrm{E}_R[R_j/s_j] = 1$, its expectation over label sampling is $\sum_j w_j y_j = \theta$, regardless of the classifier's errors. Verified labels must measure $y_j$ correctly. Retraining the proxy on the same sampled labels requires a separate argument; the fixed-proxy proof does not cover it.

Under independent Bernoulli item sampling,

$$\mathrm{Var}_R(\widetilde{\theta}) = \sum_j \frac{1 - s_j}{s_j} w_j^2 (y_j - q_j)^2. \tag{B2}$$

The realized sample gives an unbiased variance estimator by replacing each summand with $R_j(1 - s_j)w_j^2(y_j - q_j)^2/s_j^2$. Fixed-size or balanced designs require their joint-inclusion terms or a justified approximation. An inaccurate proxy can increase variance relative to omitting it, but does not introduce design bias under these assumptions. This is uncertainty from randomized label acquisition, with $y$ and $q$ fixed; it is distinct from the prediction-error process in equation (D4). Both calculations preserve the reuse of an item label across downstream records.

# C Auditing an aggregated regression covariate

The general plug-in error is $F(Cq, Z) - F(Cy, Z)$. If scalar $F$ is differentiable throughout the segment from $Cy$ to $Cq$, it also has the exact representation

$$F(Cq, Z) - F(Cy, Z) = \int_0^1 \left[ C^\mathsf{T} \nabla_T F\{Cy + sC(q - y), Z\} \right]^\mathsf{T} (q - y)\, ds. \tag{C1}$$

Unlike the fixed weights of a linear target, these effective item weights depend on unknown labels along the path. Thresholds need not be differentiable.

For the OLS coefficient in equation (3), set $d = C(q - y)$, $t = M_V C y$, $h = M_V d$, and $u = M_V Y - \beta(y)t$, so $t^\mathsf{T} u = 0$. If the predicted-regressor denominator is positive, the exact error is

$$\beta(q) - \beta(y) = \frac{h^\mathsf{T} u - \beta(y)\{t^\mathsf{T} h + h^\mathsf{T} h\}}{(t + h)^\mathsf{T}(t + h)}. \tag{C2}$$

The quadratic term can remain when first-order terms cancel, and the coefficient can move in either direction.

The conventional OLS score makes the two channels explicit. Write $a = C^\mathsf{T} M_V Y$, $G = C^\mathsf{T} M_V C$, and $U(z; \beta) = z^\mathsf{T} a - \beta z^\mathsf{T} G z$. For the fixed-sample complete-label coefficient $\beta^* = \beta(y)$, $U(y; \beta^*) = 0$, whereas

$$U(q; \beta^*) = e^\mathsf{T}(a - 2\beta^* G y) - \beta^* e^\mathsf{T} G e, \qquad e = q - y. \tag{C3}$$

If repeated predictions are meaningful while $y, C, V, Y$ remain fixed, let $b = \mathrm{E}[e]$ and $\Sigma = \mathrm{Cov}(e)$. Then

$$\mathrm{E}[U(q; \beta^*)] = b^\mathsf{T}(a - 2\beta^* G y) - \beta^*\{b^\mathsf{T} G b + \mathrm{tr}(G\Sigma)\}. \tag{C4}$$

The trace term equals $\mathrm{tr}(M_V C \Sigma C^\mathsf{T})$. With independent item errors it is $\sum_j v_j \|M_V C_{\cdot j}\|^2$: each item's contribution depends on variation in its unit-level loading after removing the controls. It is nonnegative, but the other terms in equation (C4) can have either sign, and an expected score is not the expectation of the OLS coefficient. In particular, ordinary attenuation does not follow. For fixed binary $y$ and hard binary $q$, nontrivial errors cannot have $b = 0$ item by item: a negative item

can only have a positive error, and a positive item can only have a negative one. Mean-zero error is thus not a harmless default for this prediction process.

For several covariates built from the same predictions, let $T_r = C^{(r)}y$ and $d_r = C^{(r)}e$ on the same regression rows. The $(r, s)$ error cross-product in their residualized normal equations has expectation

$$\mathrm{E}[d_r^{\mathsf{T}} M_V d_s] = b^{\mathsf{T}} C^{(r)\mathsf{T}} M_V C^{(s)} b + \mathrm{tr}\{C^{(r)\mathsf{T}} M_V C^{(s)} \Sigma\}. \tag{C5}$$

Even if distinct item errors are independent, the trace term contains $\sum_j v_j (M_V C^{(r)}_{\cdot j})^{\mathsf{T}} (M_V C^{(s)}_{\cdot j})$. One item prediction can therefore distort the cross-product between two constructed regressors. If the outcome also uses these predictions, its measurement error contributes the analogous term $\mathrm{E}[d_r^{\mathsf{T}} M_V d_0]$ to the regression numerator, with $C^{(0)}$ building the outcome. Shared errors can then create an apparent association as well as distort a regressor's denominator. These terms depend on how each aggregation weights the same item; they need not produce scalar attenuation of any coefficient.

More generally, let $D(y)$ collect $p$ columns of the form $C^{(r)}y$. With an observed outcome $Y$, the complete-label coefficient vector is

$$\boldsymbol{\beta}(y) = B(y)^{-1} m(y), \qquad m_r(y) = y^{\mathsf{T}} C^{(r)\mathsf{T}} M_V Y, \qquad B_{rs}(y) = y^{\mathsf{T}} C^{(r)\mathsf{T}} M_V C^{(s)} y, \tag{C6}$$

when $B(y)$ is nonsingular. The same item label enters every component of $m$ and $B$, so their audit uncertainty is joint. If $Y$ is also built from these labels, its numerator moments are quadratic too. This is the usual regression normal equation with its item-level construction made explicit; it does not prescribe one particular correction method.

The fixed-sample coefficient in equation (3) has a linear numerator and a quadratic denominator in the item labels. Put $r = y - q$. Then

$$\begin{aligned} N &= y^{\mathsf{T}} a = q^{\mathsf{T}} a + r^{\mathsf{T}} a, \\ D &= y^{\mathsf{T}} G y = q^{\mathsf{T}} G q + 2 r^{\mathsf{T}} G q + r^{\mathsf{T}} G r. \end{aligned} \tag{C7}$$

Let $R_j$ indicate an item's inclusion in a probability audit, with $\pi_j = \Pr_R(R_j = 1) > 0$ and $\pi_{j\ell} = \Pr_R(R_j = R_\ell = 1) > 0$ for every relevant pair. Set $\pi_{jj} = \pi_j$. The following estimates are design unbiased for the two moments, conditional on $C, V, Y, y, q$:

$$\begin{aligned} \widehat{N} &= q^{\mathsf{T}} a + \sum_j \frac{R_j}{\pi_j} r_j a_j, \\ \widehat{D} &= q^{\mathsf{T}} G q + 2 \sum_j \frac{R_j}{\pi_j} r_j (Gq)_j + \sum_{j,\ell} \frac{R_j R_\ell}{\pi_{j\ell}} r_j G_{j\ell} r_\ell. \end{aligned} \tag{C8}$$

The identities follow from $\mathrm{E}_R[R_j/\pi_j] = 1$ and $\mathrm{E}_R[R_j R_\ell/\pi_{j\ell}] = 1$. No independent-item-error or calibrated-score assumption is used. The ratio $\widehat{N}/\widehat{D}$ is generally not design unbiased and can be unstable or undefined when the estimated denominator is near zero or nonpositive. Valid inference for the ratio requires additional design analysis; its variance involves inclusion probabilities for sets of up to four items. The formulas identify the information an item audit would have to preserve. The linear-label difference estimator in equation (B1) does not by itself correct a regression that uses $Cy$ as a covariate.

For a retrospective standard error, treat $(\widehat{N}, \widehat{D})$ as a joint vector of item- and item-pair Horvitz–Thompson totals. If the required inclusion probabilities for sets of up to four items are known and positive, its design covariance can be estimated from the realized audit. When $D$ is bounded away from zero and a normal approximation for the joint moments is justified, the delta-method variance

for $N/D$ uses gradient $(1/D, -N/D^2)^\mathsf{T}$. Near a zero denominator, a ratio confidence set must instead respect the joint uncertainty in numerator and denominator; the plug-in delta standard error is unreliable. With several constructed regressors, the corresponding moment vector includes all residualized regressor–outcome and regressor–regressor cross-products, including equation (C5). Its covariance and the regression map must be propagated jointly. These are routes to inference under an observed probability-validation design, not a claim that an arbitrary validation set supports an automatic corrected standard error.

# D Repeated-prediction error and dependence

For a fixed fitted classifier and fixed labels, $\Delta$ is fixed. To discuss bias and variance, specify a repeated-prediction process, such as repeated training samples, while fixing $y, w$ and the inputs that determine $w$. All moments below condition on this fixed design. Define $b = \mathrm{E}[e]$ and the full joint covariance $\Sigma = \mathrm{Cov}(e)$, without assuming independence across items. Then

$$\begin{aligned}
\mathrm{Bias}(\widehat{\theta}) &= w^{\mathsf{T}} b, \\
\mathrm{Var}(\Delta) &= w^{\mathsf{T}} \Sigma w = \sum_j w_j^2 \Sigma_{jj} + 2 \sum_{j<\ell} w_j w_\ell \Sigma_{j\ell}, \qquad \text{(D1)} \\
\mathrm{E}[\Delta^2] &= (w^{\mathsf{T}} b)^2 + w^{\mathsf{T}} \Sigma w. \qquad \text{(D2)}
\end{aligned}$$

Equations (1)–(D2) connect the research quantity to its item weights, systematic error, and uncertainty. Covariance between item errors enters the estimate's variance. Dependence alone does not determine bias; systematic errors on items with nonzero weights can make $w^{\mathsf{T}} b$ nonzero.

Unit aggregates and regression vectors are corresponding joint quantities. Writing $\widehat{T} = Cq$, $d = \widehat{T} - T$, and $\widehat{\beta} = ACq$ gives

$$\begin{aligned}
d &= Ce, \qquad \text{(D3)} \\
\mathrm{E}[d] &= Cb, \qquad \mathrm{Cov}(d) = C\Sigma C^{\mathsf{T}}, \qquad \text{(D4)} \\
\mathrm{E}[\widehat{\beta} - \beta^*] &= ACb, \qquad \mathrm{Cov}(\widehat{\beta} - \beta^*) = AC\Sigma C^{\mathsf{T}} A^{\mathsf{T}}. \qquad \text{(D5)}
\end{aligned}$$

The same item error is reused across every unit linked to that item. These formulas quantify the specified prediction-error process; they do not include errors in the item-to-unit map, unit sampling, or uncertainty about the construct. They concern a label-derived aggregate as an outcome. Using that aggregate as a regressor is a different errors-in-variables problem treated in Section 2.

## D.1 Shared labels and correlated item errors

These are separate mechanisms under a repeated-prediction model. First, a single item error can appear in several units' aggregates. Even if the item errors $e_j$ are independent across $j$, units that share an uncertain item can have correlated aggregate errors. This is a standard common-shock structure, analogous to shift-share inference (Adão et al., 2019); it does not assert correlation between distinct item errors. Second, distinct item errors may themselves covary under repeated fitting. Shared training data or model parameters can produce such covariance, but using one fitted classifier does not by itself establish it. Units linked to disjoint items can have correlated aggregate errors only under this second mechanism or another shared source of uncertainty. In equation (D1), positively correlated item errors increase variance when their weights have the same sign, and decrease it when their weights have opposite signs.

For a limiting example, suppose centered errors are shared within item families: $e_j - b_j = u_{g(j)}$, with independent family shocks $u_g$ of variance $\tau_g^2$. This is compatible with binary predictions when the classifier assigns one common random prediction to every item in a family. Then

$$\mathrm{Var}(\Delta) = \sum_g \tau_g^2 \left( \sum_{j:g(j)=g} w_j \right)^2. \qquad \text{(D6)}$$

A family of $m$ items with unit weights contributes $m^2 \tau_g^2$, whereas assuming independent item errors would give $m\tau_g^2$. Signed weights can instead cancel within a family. The relevant concentration is in

aggregate family influence, not just individual item influence. This example is a special covariance structure, not an assertion that the right families are known or that all within-family mistakes are perfectly correlated.

Independent item errors are another special case: $\Sigma$ is diagonal, with entries $v_j$. For any unit aggregate $T = Cy$, equation (D4) reduces to

$$\mathrm{Var}(d_i) = \sum_j c_{ij}^2 v_j, \qquad \mathrm{Cov}(d_i, d_\ell) = \sum_j c_{ij} c_{\ell j} v_j. \tag{D7}$$

Let $\alpha_j$ be the false-positive probability on a truly negative item and $\beta_j$ the false-negative probability on a truly positive item. For binary predictions, $v_j = \alpha_j(1-\alpha_j)$ on negative items and $v_j = \beta_j(1-\beta_j)$ on positive items. In browsing, repeated visits multiply one classification error; treating visits as independent would incorrectly replace squared visit weights by visit counts. More generally, any reuse of an uncertain item can induce dependence across units.

The same predictions can build several different aggregates. If the $r$th aggregate uses a fixed matrix $C^{(r)}$, then

$$d^{(r)} = C^{(r)} e, \qquad \mathrm{Cov}\{d^{(r)}, d^{(s)}\} = C^{(r)} \Sigma C^{(s)\mathsf{T}}. \tag{D8}$$

For scalar estimates with item-weight vectors $w^{(r)}$, independent item errors yield $\mathrm{Cov}(\Delta_r, \Delta_s) = \sum_j v_j w_j^{(r)} w_j^{(s)}$. Thus each item contributes a rank-one term to the joint covariance of all estimates. A heavily weighted item can move several reported quantities together; subtracting two of them can cancel its error. For a sequence of larger unit samples, independent item errors with $v_j\{w_j^{(r)}\}^2$ bounded away from zero for even one item prevent the prediction-error variance of that estimate from vanishing. More units alone do not supply more independent predictions for that item. The covariance statements require a repeated-prediction model; $d^{(r)} = C^{(r)} e$ remains an exact identity for fixed realized predictions.

For a fixed linear unit contrast $a^\mathsf{T} d$, equal item-error variances give a simple overlap diagnostic:

$$R(a, C) = \frac{\sum_j (\sum_i a_i c_{ij})^2}{\sum_i a_i^2 \sum_j c_{ij}^2}. \tag{D9}$$

The numerator is its variance, apart from the common variance factor. The denominator is what that variance would be if each unit's aggregate error were independent while retaining its marginal variance. Thus $R$ measures the effect of discarding covariance across units for this one prediction-error component. It is at least one for nonnegative weights and counts, but may be above or below one for a signed contrast. It is not a general standard-error correction: unequal item variances, correlated item errors, unit sampling, and other errors require their own treatment.

An uncertainty procedure must follow the same structure as the estimand. Resampling downstream records while holding predictions fixed does not propagate repeated-training uncertainty. Each replicate must preserve reused labels across records and the joint distribution of errors across items. Independently perturbing each item label imposes a diagonal $\Sigma$; resampling the classifier fit and predicting all items together can preserve dependence induced by that fitting process, provided the resampling scheme represents that process. A prediction-error covariance cannot simply be added to an arbitrary reported regression covariance: the components may overlap or be dependent. Neither larger downstream samples nor clustering records alone guarantees that this error disappears, and standard errors do not repair bias.

### D.2 Many items do not ensure a normal approximation

Even with independent item errors, diffuse variance shares alone do not justify Gaussian inference. Let all true labels be zero and let each of $k$ items be falsely labeled positive independently with

probability $1/k$. With $w_j = 1$, total error $B_k$ is binomial with mean one and variance $1 - 1/k$. The largest item variance share is $1/k$, yet $B_k - 1 \Rightarrow \text{Poisson}(1) - 1$, not a normal distribution. At $k = 10{,}000$, an oracle interval using the exact bias and variance but a normal 1.96 cutoff has coverage 91.97%. This calculation uses exact binomial probabilities, not simulation.

For independent bounded binary errors, a sufficient Lindeberg condition is

$$\frac{\max_j |w_j|}{\sqrt{\sum_j w_j^2 v_j}} \to 0.$$

A small maximum variance share $\max_j w_j^2 v_j / \sum_j w_j^2 v_j$ is weaker when errors become rare. The example violates the former condition and satisfies the latter. A growing number of items cannot substitute for a condition on the distribution of their errors. This is a prediction-error example, distinct from the fixed-label audit design.

## E Count-certificate construction and audit diagnostics

Partition items into strata $h = 1, \dots, H$ before observing their audited labels. Sample $n_h$ of the $M_h$ items without replacement. If $K_h$ is the unknown number of incorrect predictions and $X_h$ the sampled error count, then $X_h \sim \text{Hypergeometric}(M_h, K_h, n_h)$. Assign nonnegative error budgets $\delta_h$ with $\sum_h \delta_h \le \delta$. Retain integers $K$ for which both $\Pr_K(X_h \le x_h) \ge \delta_h/2$ and $\Pr_K(X_h \ge x_h) \ge \delta_h/2$. Their extrema are $\ell_h, u_h$. A census stratum has $\ell_h = u_h = x_h$; an unaudited stratum contributes no count restriction.

Let $O$ be the audited items, $U_h$ the unaudited items in stratum $h$, $c_j = (2q_j - 1)w_j$, and $d_O = \sum_{j \in O} w_j(q_j - y_j)$. The smallest and largest compatible errors are

$$L_A = d_O + \sum_h \min_{\ell_h - x_h \le m \le u_h - x_h} S_-(c_{U_h}, m),$$
$$U_A = d_O + \sum_h \max_{\ell_h - x_h \le m \le u_h - x_h} S_+(c_{U_h}, m). \tag{E1}$$

The optimization is over integers. With signed weights, an extreme need not occur at the largest admissible error count.

*Proof of Proposition 3.* Inverting each pair of hypergeometric tails excludes the true $K_h$ with probability at most $\delta_h$. A union bound therefore includes all true counts with probability at least $1-\delta$. On that event, the true errors in $U_h$ number between $\ell_h - x_h$ and $u_h - x_h$. Sorting and choosing a count in that range gives the extrema in equation (E1); the choices are independent restrictions across strata and hence jointly attainable. The true label vector lies in the same configuration set for every $w$, so the coverage event applies simultaneously to all its linear contrasts. □

The labels and predictions are fixed under this design. Systematic or correlated classifier mistakes do not invalidate coverage, which is over repeated audit samples. The count region can be conservative, and sharp projection of that region does not make its interval shortest. The guarantee concerns one label vector; combining separate audits of several outcomes needs an additional error allocation. Optional stopping, adaptive strata, and imperfect audit labels require other procedures or assumptions.

**Information lost by count projection.** Let all $N$ predictions equal one, half the weights equal $1/N$ and half equal $-1/N$. Suppose a fraction $p < 1/2$ of labels in each half are wrong. The true contrast is zero. If $n \to \infty$ but $n/N \to 0$, an unstratified audit consistently learns the overall error fraction. Yet unaudited errors can still be placed almost entirely on either weight sign: the projected interval width converges to $2p$. A direct estimate of the weighted audit residual instead has standard error of order $n^{-1/2}$. Separating the two weight signs into strata removes this particular information loss.

For a declared contrast, the stratified difference estimator uses $z_j = w_j(y_j - q_j)$ directly:

$$\widetilde{\theta} = \widehat{\theta} + \sum_h \frac{M_h}{n_h} \sum_{j \in O_h} z_j, \qquad \widehat{V} = \sum_h \frac{M_h^2(1 - n_h/M_h)}{n_h} s_{z,h}^2. \tag{E2}$$

Under independent stratum draws, the estimate and variance estimator are design unbiased when each noncensus stratum has at least two sampled items. Normal intervals need a separate approximation. A conservative finite-sample alternative applies Hoeffding's without-replacement inequality (Hoeffding, 1963) to each bounded residual mean and sums the stratum radii. More efficient weighted audit procedures exist (Shekhar et al., 2023).

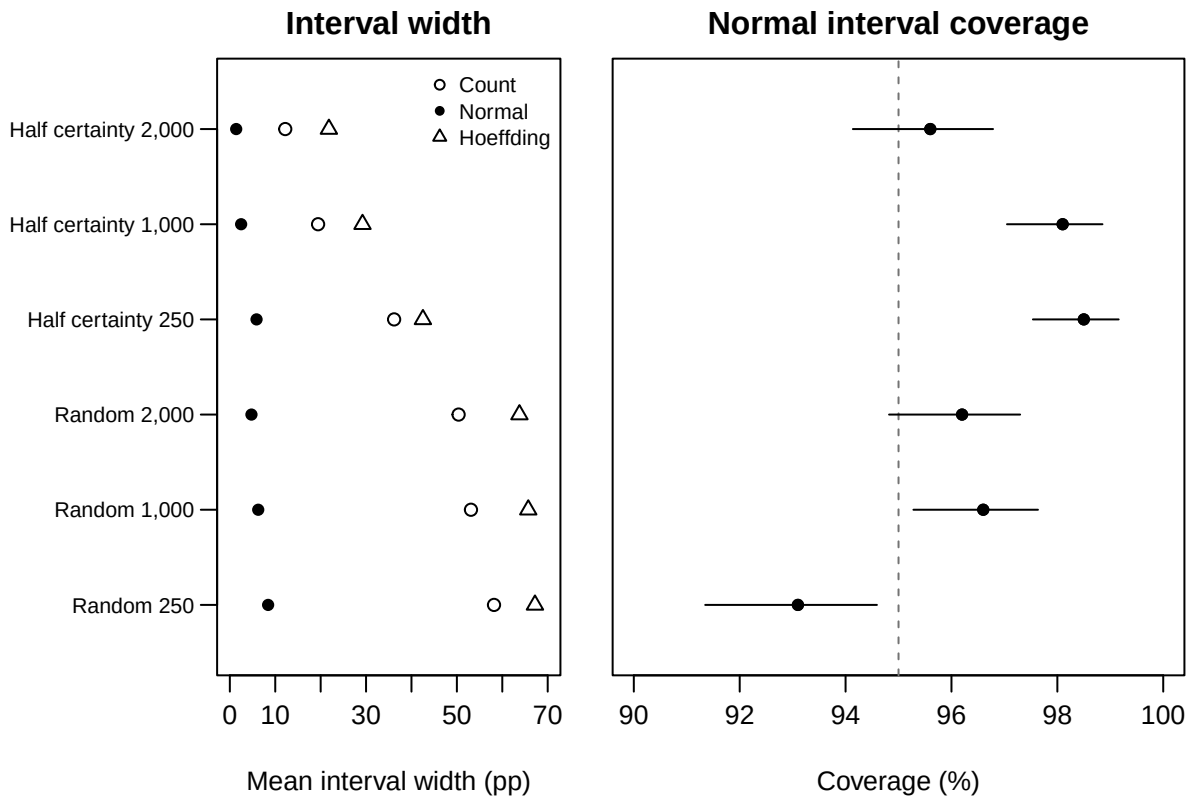



Figure E1: Width and coverage for the Facebook-pixel female share coefficient. Rows compare audit budgets and designs in the fixed browsing frame. Left: mean interval width in percentage points over 1,000 audits. Right: normal interval coverage, with exact 95% binomial Monte Carlo intervals and a dashed 95% reference. Count and Hoeffding intervals covered in every simulated draw. "Half certainty" audits the largest absolute influences with half the budget. Count intervals cover all contrasts of this outcome simultaneously; direct intervals target this coefficient. The randomness is domain auditing; users and predictions are fixed.

Table E1: All audit designs and methods for the focal coefficient

| Design | Budget | Interval | Coverage (%) | Width (pp; MCSE) | Correct sign (%) | Wrong sign (%) |
|---|---|---|---|---|---|---|
| Random | 250 | Count projection | 100.0 | 58.2 (0.06) | 0.0 | 0.0 |
| Random | 250 | Direct normal | 93.1 | 8.4 (0.50) | 39.6 | 0.0 |
| Random | 250 | Direct Hoeffding | 100.0 | 67.2 (0.01) | 0.0 | 0.0 |
| Random | 1,000 | Count projection | 100.0 | 53.1 (0.05) | 0.0 | 0.0 |
| Random | 1,000 | Direct normal | 96.6 | 6.3 (0.19) | 32.9 | 0.0 |
| Random | 1,000 | Direct Hoeffding | 100.0 | 65.7 (0.02) | 0.0 | 0.0 |
| Random | 2,000 | Count projection | 100.0 | 50.4 (0.05) | 0.0 | 0.0 |
| Random | 2,000 | Direct normal | 96.2 | 4.8 (0.11) | 34.4 | 0.0 |
| Random | 2,000 | Direct Hoeffding | 100.0 | 63.8 (0.03) | 0.0 | 0.0 |
| Half certainty | 250 | Count projection | 100.0 | 36.2 (0.05) | 0.0 | 0.0 |
| Half certainty | 250 | Direct normal | 98.5 | 5.9 (0.29) | 40.6 | 0.0 |
| Half certainty | 250 | Direct Hoeffding | 100.0 | 42.5 (0.00) | 0.0 | 0.0 |
| Half certainty | 1,000 | Count projection | 100.0 | 19.4 (0.04) | 0.0 | 0.0 |
| Half certainty | 1,000 | Direct normal | 98.1 | 2.5 (0.04) | 41.9 | 0.0 |
| Half certainty | 1,000 | Direct Hoeffding | 100.0 | 29.2 (0.00) | 0.0 | 0.0 |
| Half certainty | 2,000 | Count projection | 100.0 | 12.2 (0.03) | 0.0 | 0.0 |
| Half certainty | 2,000 | Direct normal | 95.6 | 1.4 (0.01) | 65.0 | 0.0 |
| Half certainty | 2,000 | Direct Hoeffding | 100.0 | 21.8 (0.00) | 0.0 | 0.0 |

Each row summarizes 1,000 domain audits. Width is the mean in percentage points, with its Monte Carlo standard error in parentheses. Sign columns report exclusion of zero in the correct or wrong direction. All methods within a design use the same draws. Exact binomial Monte Carlo intervals and replicate results accompany the package.

The random and half-certainty designs were chosen for one coefficient. They do not establish that this design is efficient for other contrasts. The count certificate covered the target in every simulated draw but never excluded zero. Direct normal intervals were much narrower, yet coverage in the smallest random design fell below the nominal level. The finite-sample Hoeffding baseline was wider than count projection. These comparisons do not identify a finite-sample-valid, efficient direct interval for this design.

Table E2: Refining validation strata for the focal coefficient

| Strata | Exact-count width (pp) | Audit width (pp; MCSE) | Coverage (%) |
|---|---|---|---|
| 2 | 49.7 | 53.2 (0.05) | 100.0 |
| 10 | 34.7 | 47.1 (0.05) | 100.0 |
| 50 | 21.8 | 48.2 (0.03) | 100.0 |

Rank bins of signed influence cross predicted labels. Exact-count width conditions on the population count in every stratum; audit width averages 95% count-projection intervals over 1,000 domain samples of size 1,000. Width uncertainty is its Monte Carlo standard error. These partitions were examined after the initial audit results. No design certified the sign.

The narrower exact-count range from finer strata need not translate into a narrower audit interval: the fixed sample is divided among more strata and the error budget is split across their discrete hypergeometric intervals. The 10- versus 50-stratum mean-width difference should be read with the reported Monte Carlo uncertainty, not as evidence of an optimal partition.

# F Retaining the highest-visit domain labels

Table F1 repeats the count-information exercise after retaining both observed classification labels for the highest-visit domains. For the remaining domains it retains only the numbers positive under Piedomains alone and under the provider alone. The ordering uses visit count with domain name to break ties. The count range is sharp under exactly this information: the head contribution is fixed, while the tail discrepancies can occupy any compatible tail domains. At zero retained domains it equals the range in Table 3. The provider's observed visit share lies inside every range.

Table F1: Count-compatible provider visit shares after retaining label pairs for the highest-visit domains

| Provider | Domains retained | Visits retained (%) | Count range (%) | Width (pp) |
|---|---|---|---|---|
| Bitdefender | 0 | 0.0 | [0.07, 82.11] | 82.05 |
| Bitdefender | 10 | 42.2 | [0.07, 40.06] | 39.99 |
| Bitdefender | 50 | 56.8 | [0.07, 25.80] | 25.73 |
| Bitdefender | 250 | 71.5 | [1.00, 13.65] | 12.64 |
| Bitdefender | 1,000 | 83.4 | [1.33, 5.98] | 4.64 |
| Forcepoint | 0 | 0.0 | [0.07, 80.61] | 80.54 |
| Forcepoint | 10 | 40.6 | [0.07, 40.16] | 40.09 |
| Forcepoint | 50 | 55.1 | [0.07, 25.99] | 25.92 |
| Forcepoint | 250 | 69.6 | [0.86, 14.01] | 13.15 |
| Forcepoint | 1,000 | 81.9 | [1.24, 6.24] | 4.99 |
| alphaMountain | 0 | 0.0 | [0.04, 68.07] | 68.03 |
| alphaMountain | 10 | 30.1 | [0.04, 38.33] | 38.30 |
| alphaMountain | 50 | 49.0 | [0.56, 21.37] | 20.82 |
| alphaMountain | 250 | 67.2 | [1.44, 9.12] | 7.68 |
| alphaMountain | 1,000 | 81.5 | [1.87, 4.03] | 2.16 |

The percentages use each provider row's categorized-domain visit total as its denominator. "Visits retained" is the share of that total on domains whose two observed labels remain known. Width is in percentage points. The providers use different category definitions and neither label is an adjudicated content truth. Retaining a label pair here is an information exercise, not a claim that a manual audit was performed.